

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

1 Above Town	4.8	4.5	5.2	5.0	4.9	5.1	5.2
2 Below Town	5.0	4.7	4.9	5.0	4.8	4.9	5.1

- Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.
- Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

9.

	2	1	2	1	2	2	1	2	1	1
data	4.7	4.8	4.8	4.9	4.9	4.9	5	5	5.1	5.2
rank	1	2.5	2.5	5	5	5	7.5	7.5	9	10

H_0 : two population distribution are the same.

H_a : Population 1 is to the right of the population 2.

$$n_1 = n_2 = 5 \quad T_1 = 2.5 + 5 + 7.5 + 9 + 10 = 34$$

$$T_1^* = n_1(n_1 + n_2 + 1) - T_1 = 5 \times 11 - 34 = 21 \quad T = T^* = 21$$

$$\alpha = 0.05 \quad T = 19$$

$$RR: \{T, T \leq 19\}$$

$$21 > 19 \Rightarrow \text{non-reject } H_0$$

There isn't sufficient evidence to say that population 1 distribution is to the right of population 2 distribution.

$$b. \text{ Above} = X_1 = \frac{4.8 + 4.5 + 5.2 + 5.0 + 4.9 + 5.1}{5} = 4.9$$

$$S_1^2 = 0.025$$

$$\text{Below} = X_2 = \frac{5.0 + 4.7 + 4.9 + 4.8 + 4.9}{5} = 4.86$$

$$S_2^2 = 0.013$$

$$\frac{\text{larger } S^2}{\text{smaller } S^2} = \frac{0.025}{0.013} = 1.923 < 3$$

$$\Rightarrow \text{assume } \sigma_1^2 = \sigma_2^2$$

$$> 2 \quad (n_1 - 1)S_1^2 + (n_2 - 1)S_2^2$$